System Validation Report

Project: Traffic Management System (Simulation)
Course: Software Engineering UE24CS341A, PES University
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Tested revision: `main` @ 089bf3a plus the test files and evidence added by this PR. No application source file is changed by this PR.
Reference documents: Software Test Plan v1.0, SAD v1.0, SRS v1.0

1. Scope and method

This report records the validation of the Traffic Management System (Simulation) against the Software Test Plan. Testing was carried out on the integrated build of the `main` branch, which includes the Simulation Engine, Queue, Signal, Vehicle and Statistics managers, the Console Interface, the Input Validator and the File Manager.

Item	Value
Environment	Ubuntu 24.04 (WSL2), Windows 11 host
Compiler	g++ 13.3.0, C++17, flags `-std=c++17 -g -O0 -Wall -Wextra`
Memory tool	Valgrind 3.22.0 (`--leak-check=full --track-origins=yes`)
Coverage tool	gcovr 7.0 / gcov, `src/` only (main.cpp excluded)
Unit test method	each `tests/test_*.cpp` is compiled with all `src/*.cpp` except `main.cpp` and run
System test method	scripted console sessions and driven `ConsoleInterface` sessions

2. Summary of results

Measure	Result
Unit test executables	13 run (includes `test_stress_100`), 0 failures. Unit test report, Valgrind unit report and this report all count the same 13 programs.
Line coverage	**96.86%** (525 of 542 lines)
Branch coverage	**90.87%** (428 of 471 branches)
Valgrind	0 errors, no leaks on all 12 unit test programs (`docs/valgrind_unit_tests.txt`) and on `test_stress_100` plus three scripted runs of `traffic_sim` (`docs/valgrind_full_program.txt`)
Conflicting signal states observed	0 (TC-SAFE-01, 200 ticks plus manual mode)
Open Critical or High defects	0 (three Low-severity items are open, see section 5)

3. Test case results

TC ID	Requirement	Scenario	How tested	Expected	Actual	Result
TC-VM-01..05	TMS-F-001..005	Vehicle creation, unique IDs, approach assignment, auto-generation	`test_managers`, `test_vehicle_generation`, `test_console_actions`	IDs unique and increasing; correct approach and entry tick; probability 0 gives none, 1 gives always; invalid probability rejected	All checks passed	Pass

TC-Q-01..05	TMS-F-006..010	FIFO per approach, empty-queue handling, live counts, reset	`test_queue_manager`, `test_managers`	FIFO order kept; dequeue on empty returns false; counts correct; reset clears queues	All checks passed	Pass
TC-SIG-01	TMS-F-011	Signal states at start	`test_managers`	N+S green, E+W red at tick 0	As expected	Pass
TC-SIG-02	TMS-F-012	Automatic safe phase cycle	`test_managers`	Green, yellow, then other axis green, cycle repeats	As expected	Pass
TC-SIG-03	TMS-F-013	Configurable durations, invalid values ignored	`test_managers`, `test_console_actions`	New durations applied; invalid values rejected	As expected	Pass
TC-SIG-04	TMS-F-014, F-015	Manual signal control, automatic resume	`test_managers`, `test_console_actions`	One approach green in manual mode; automatic cycle resumes	As expected	Pass
TC-SAFE-01	TMS-NF-006	No conflicting greens	`test_managers`	No conflicting signals over 200 ticks and in manual mode	0 conflicts	Pass
TC-SIM-01..06	TMS-F-016..021	Tick loop; Start, Pause, Resume, Reset, Stop; actions in wrong state	`test_simulation_engine`, `test_console_actions`, `test_console_interface`	Clock advances only when running; frozen when paused; Reset clears clock, queues, statistics; invalid actions rejected without crash	All checks passed	Pass
TC-SIM-07	TMS-F-020	Vehicle IDs restart at 1 after Reset	Create vehicles, Reset, create a vehicle (manual check against `main`)	New vehicle has ID 1	IDs keep increasing after Reset; `SimulationEngine::reset()` does not reset the vehicle counter	**Fail (open, Low)**, see observation 3
TC-MET-01..05	TMS-F-022..026	Waiting and processed counts, average wait, throughput, summary	`test_statistics_manager`, `test_statistics_summary`	Average wait = total wait / processed; zero-data safe; negative wait clamped to 0; summary printed	All checks passed	Pass
TC-DATA-01..03	TMS-F-027..029	Save and load settings, export CSV, malformed file	`test_file_manager`, `test_console_files`	Valid files saved and loaded; CSV created; missing or malformed file rejected, no crash	All checks passed	Pass
TC-SEC-01	TMS-SR-001	Input validation (menu, integers, decimals)	`test_input_validator`, `test_input_validator_double`, `test_console_actions`	Non-numeric and out-of-range input rejected with message	All checks passed	Pass
TC-SEC-02	TMS-SR-002	Empty-queue dequeue	`test_queue_manager`, `test_simulation_engine`	Returns false, no invalid access	As expected	Pass
TC-SEC-03	TMS-SR-003	Configuration file validation	`test_file_manager`, `test_console_files`	Malformed and missing files rejected	As expected	Pass
TC-SEC-04	TMS-SR-004	Unexpected input and end of input	`test_console_interface`	Clean exit, no abnormal termination	As expected	Pass
TC-SEC-05	TMS-SR-005	Memory de-allocation	Valgrind on all unit tests	No leaks, no invalid accesses	0 errors, all heap blocks freed	Pass
TC-PERF-01	TMS-NF-001	100 active vehicles, no noticeable delay	`tests/test_stress_100.cpp` (100 queued vehicles; 100 auto-generated arrivals using the same generate-and-tick loop as the console) and the real program with `tests/stress_input.txt` and `tests/autogen_input.txt`	All 100 processed, no conflicting greens, unique IDs, no noticeable

delay | 100 processed in 70 ticks in about 0.1 ms; both console runs end with 100 processed and 0 waiting; see `docs/stress_test_output.txt`, `docs/autogen_run_output.txt` | Pass |
TC-MEM-01	TMS-NF-007	Memory safety, full program under load	Valgrind on `test_stress_100` and on `traffic_sim` with `stress_input.txt`, `edge_input.txt`, `autogen_input.txt`	0 errors, no leaks	All four runs: exit 0, all heap blocks freed, 0 errors; see `docs/valgrind_full_program.txt`	Pass
TC-REL-01	TMS-NF-002	Reproducibility with a fixed seed	Review of `VehicleManager::generateVehicle`	Same seed gives identical runs	Generation uses `std::random_device`; no way to set a seed, so the requirement cannot be tested	**Not passed (outstanding)**, see observation 2
TC-UX-01	TMS-NF-003	Console usability	Manual walkthrough of the 15-option menu with valid and invalid input (add vehicles, start, step, view, statistics, durations, manual signals, stop)	Clear menu, status line after every action, clear error messages with re-prompt	Status line (state, tick, signal mode, auto-generation) shown after every action; invalid input such as `abc` or `99` gives "Invalid input. Enter a whole number between 0 and 15." and re-prompts; per-tick line shows signals, queue sizes and each vehicle that crossed with its wait; statistics shows ticks, processed, average wait, throughput	Pass
TC-MAINT-01	TMS-NF-004	Maintainability	Code review of module boundaries and build warnings	One responsibility per module; clean build	9 source files with one responsibility each (largest: ConsoleInterface.cpp, 459 lines); main.cpp is 26 lines; build with `-Wall -Wextra` gives 0 warnings	Pass
TC-PORT-01	TMS-NF-005	Windows and Linux build	Makefile on Ubuntu (WSL) and GitHub Actions CI; Windows build command in section 6	Builds on both	Linux: clean build, 0 warnings (Ubuntu 24.04, g++ 13.3) and CI green. Windows: builds with g++ using the command in section 6 and the program runs.	Pass on Linux and Windows

4. Code coverage

Unit tests were instrumented with `--coverage`. `main.cpp` is excluded because it only creates the engine and starts the console; the console logic in `ConsoleInterface.cpp` is included. All percentages in the reports are given to one decimal place (the gcovr tables show whole percentages; each report ends with the one-decimal total).

File	Lines	Branches
ConsoleInterface.cpp	254/268 (94.8%)	188/210 (89.5%)
FileManager.cpp	52/55 (94.5%)	78/82 (95.1%)
InputValidator.cpp	48/48 (100.0%)	65/69 (94.2%)
QueueManager.cpp	24/24 (100.0%)	12/12 (100.0%)
SignalManager.cpp	45/45 (100.0%)	28/30 (93.3%)
SimulationEngine.cpp	51/51 (100.0%)	32/37 (86.5%)
StatisticsManager.cpp	34/34 (100.0%)	6/6 (100.0%)
vehicle_manager.cpp	17/17 (100.0%)	19/25 (76.0%)
Total	**525/542 (96.86%)**	**428/471 (90.87%)**

Branch figures exclude compiler-generated exception and unreachable branches (`--exclude-throw-branches --exclude-unreachable-branches`). The remaining uncovered lines are error paths such as file-system failures and I/O stream failures that are not practical to trigger in unit tests; the uncovered branches in `vehicle_manager.cpp` are the random-number

paths. Full detail is in `docs/code_coverage_report.txt` and `docs/branch_coverage_report.txt`.

5. Observations and open items

This PR adds tests and test evidence only; it does not change application code. Three items found during testing are **open** and are reported to the team for a decision. None is marked as passed.

#	Item	Severity	Status and disposition
1	**Duplicate automatic generation.** `ConsoleInterface::maybeGenerateVehicle` (called from `doStep`) and `SimulationEngine::generateVehicleIfDue` (called from `tick`) are two separate implementations with separate enable flags and probability. The console does not call `SimulationEngine::setAutoGeneration`. The console version is what the application uses; the engine version is used only by tests.	Low (design risk, no failing test)	**Open.** Recommended fix: the console configures the engine through `setAutoGeneration` and the console copy is removed. Mitigation in this PR: the real console path is tested through the real program (`stress_input.txt`, `autogen_input.txt`) as well as through the unit tests. Owner and timing to be agreed with the team.
2	**No fixed seed (TC-REL-01, TMS-NF-002).** `VehicleManager::generateVehicle` seeds a static generator from `std::random_device` and offers no way to set a seed.	Low	**Open, TC-REL-01 not passed.** The requirement is not met. It is recorded as an outstanding requirement. Tests avoid the dependency by using probability 0.0 and 1.0. Disposition to be agreed with the team (implement a seed option, or accept the gap for this release).
3	**Vehicle IDs do not restart at 1 after Reset (TC-SIM-07).** The team asked for the restart. `SimulationEngine::reset()` does not reset the `VehicleManager` counter. IDs stay unique, so no automated test fails.	Low	**Open, TC-SIM-07 failed.** It stays open in this report, the test case document and the RTM until the behaviour is implemented and tested.

No Critical or High defects were found.

6. Conclusion

Automated tests executed: 13 test executables (12 unit test programs + test_stress_100), 0 failures; 96.86% line coverage and 90.87% branch coverage; Valgrind reported no errors or leaks on the unit programs and on test_stress_100 plus three scripted runs of traffic_sim. TC-UX-01 and TC-MAINT-01 were executed manually. Open items: TC-SIM-07 (failed, open), TC-REL-01 (partial, no seed option), duplicate auto-generation

Not passed or not complete:
- TC-REL-01 (not passed, outstanding; no seed option)(observation 2).
- TC-SIM-07 (IDs restart after Reset): failed, open (observation 3).

Windows build: g++ -std=c++17 -Wall -Wextra -Iinclude (Get-ChildItem src*.cpp).FullName -o traffic_sim_win.exe (PowerShell). Result: built without errors

7. Evidence in `docs/`

- `unit_test_report.txt`
- `code_coverage_report.txt`
- `branch_coverage_report.txt`
- `valgrind_unit_tests.txt`
- `valgrind_full_program.txt`
- `stress_test_output.txt`
- `autogen_run_output.txt`
- `system_validation_report.md` (this file)
- `test_case_document.md`